

Zero-Shot Ultra-Quality 4D-MRI Motion Reconstruction Framework Based on Augmented Digital Phantom Pre-Training

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1. Background/Motivation

- Ultra-quality 4D-MRI (UQ-4D-MRI) combines high-resolution 3D MRI with respiratory-resolved 4D MRI for abdominal motion management.
- Progress is limited by the scarcity and heterogeneity of clinical 4D-MRI data.
- Models trained on small site-specific cohorts may overfit to acquisition-specific appearance.
- Goal: develop a patient-data-free zero-shot reconstruction framework with no clinical fine-tuning.

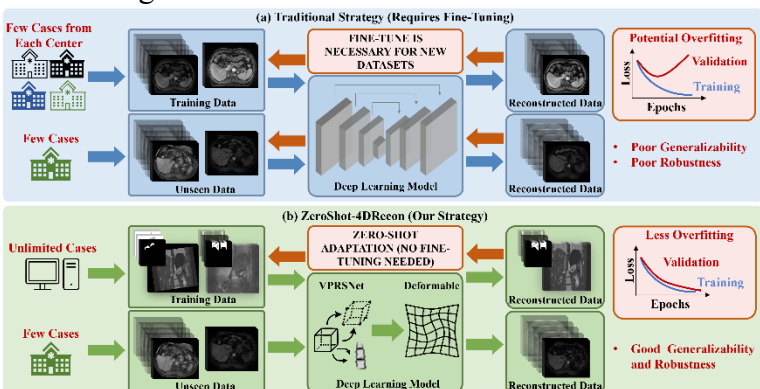


Fig. 1. Conventional patient-data fine-tuning vs. ZeroShot-4DRecon.

2. Main Contributions

- Synthetic-data pretraining**
1,000 XCAT-derived 4D phantom instances and 1,000,000 augmented training samples.
- Affine-to-deformable framework**
Vertebra-guided VPRSNet for global alignment plus anatomy-guided deformable motion estimation.
- Multi-center evaluation**
61 patients and 69 modality-specific 3D-4D MRI pairs from two centers.

3. Methods: ZeroShot-4DReconFramework

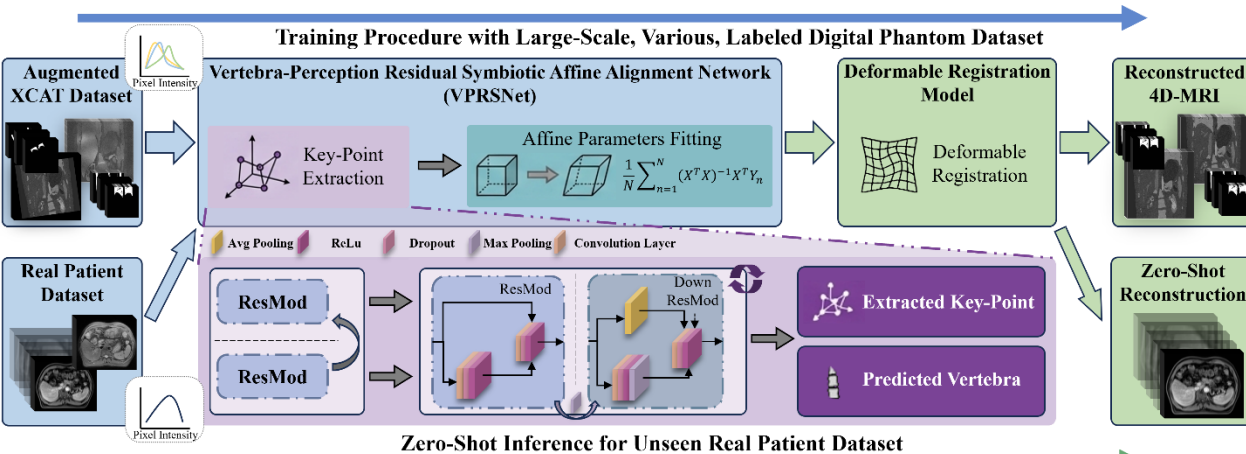


Fig. 2. Overall framework of ZeroShot-4DRecon. Augmented XCAT and real patient data are fed into VPRSNet for affine alignment, followed by deformable registration, leading to reconstructed 4D-MRI.

Stage 1: VPRSNet for Global Affine Alignment

- Estimates global affine alignment using vertebra-guided keypoint correspondence.
- Multi-scale vertebral features guide keypoint extraction toward relatively stable anatomy.
- Paired keypoints are used for least-squares affine fitting.

Stage 2: Deformable Motion Estimation

- A UNet-based deformable network estimates local respiratory motion after affine alignment.
- Anatomy-guided deformable registration recovers fine-grained motion.
- Final output: reconstructed UQ-4D-MRI.

5. Quantitative Results

Dataset / Method	Vertebra LCC ↑	NMI ↑	LCC ↑	Movement Error [mm] ↓
Center 1/T1-w (n = 53)				
JointMorph (2024)	0.180 ± 0.050	0.281 ± 0.069	0.038 ± 0.009	21.027 ± 11.641
ANTs	0.269 ± 0.072	0.347 ± 0.052	0.056 ± 0.018	6.252 ± 5.901
DDEM Framework (2022)	0.297 ± 0.043	0.389 ± 0.068	0.066 ± 0.014	4.962 ± 2.468
ZeroShot-4DRecon (Ours)	0.307 ± 0.048	0.390 ± 0.050	0.070 ± 0.019	4.812 ± 3.681
Center 2/T1-w (n = 8)				
JointMorph (2024)	0.255 ± 0.049	0.328 ± 0.071	0.103 ± 0.015	16.456 ± 4.303
ANTs	0.263 ± 0.087	0.328 ± 0.085	0.104 ± 0.017	2.284 ± 1.477
DDEM Framework (2022)	0.300 ± 0.070	0.345 ± 0.077	0.116 ± 0.016	1.509 ± 0.698
ZeroShot-4DRecon (Ours)	0.311 ± 0.082	0.360 ± 0.085	0.120 ± 0.015	1.558 ± 1.897
Center 2/T2-w (n = 8)				
JointMorph (2024)	0.217 ± 0.032	0.323 ± 0.043	0.105 ± 0.027	15.351 ± 3.316
ANTs	0.213 ± 0.040	0.323 ± 0.050	0.101 ± 0.031	2.814 ± 1.873
DDEM Framework (2022)	0.232 ± 0.033	0.329 ± 0.048	0.121 ± 0.028	2.111 ± 1.120
ZeroShot-4DRecon (Ours)	0.227 ± 0.041	0.319 ± 0.049	0.106 ± 0.028	2.485 ± 0.909

7. Conclusion /Take-Home Message

- ZeroShot-4DRecon enables patient-data-free zero-shot abdominal UQ-4D-MRI reconstruction.
- Without any clinical fine-tuning, it improved over evaluated zero-shot and iterative baselines on most metrics across the external cohorts.
- Augmented digital-phantom pretraining is a promising strategy for zero-shot abdominal UQ-4D-MRI reconstruction.
- Broader transferability across organs, pathologies, and modalities remains to be established.

4. Data and Phantom Pretraining

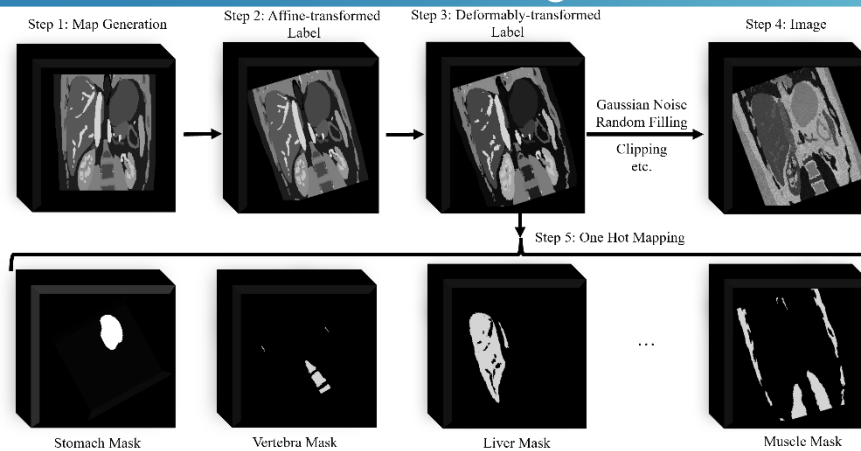


Fig. 3. Digital-phantom augmentation pipeline.

- 1,000 4D XCAT label maps sampled from a 35-dimensional physiological parameter space.
- Augmented to 1,000,000 training samples.
- Random affine perturbations: translation 0-3 px, rotation 0-3%, scale 0.97-1.03.
- Random deformable transformations: stationary-velocity-field SD 0-0.6.
- Appearance augmentation: Gaussian noise 0.01-0.1, FOV clipping 0-0.05, intensity refilling.

6. Qualitative Results

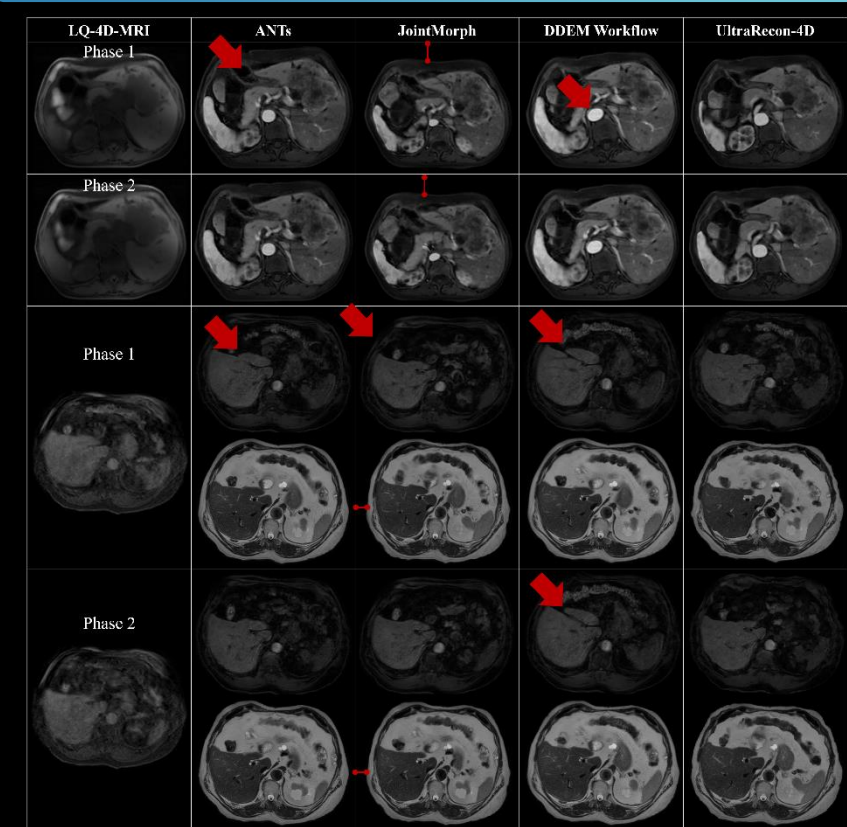


Fig. 4. Representative respiratory-motion frames from the two clinical centers.